

SAFETY DATA SHEET

MARSHAL 200SC



Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : MARSHAL 200SC

Other means of identification : CARBOSULFAN 200 G/L SC

Recommended use of the chemical and restrictions on use

Recommended use : Can be used as insecticide only.

Restrictions on use : Use as recommended by the label.

Manufacturer or supplier's details

Company : FMC Vietnam Company Limited

Address : No.12, Lot B, Thong Nhat Road
Song Than 2 Industrial Zone, Di An Ward
Di An Town, Binh Duong Province
Vietnam

Telephone : +842743790503

Telefax : +842743790501

E-mail address : SDS-Info@fmc.com

Emergency telephone : For leak, fire, spill or accident emergencies, call:
+(84)-444581938 (CHEMTREC Vietnam)

Medical emergency:
All other countries: +1 651 / 632-6793 (Collect)

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Oral) : Category 4

Acute toxicity (Inhalation) : Category 4

Specific target organ toxicity - single exposure : Category 1 (Nervous system, Bladder, Gastro-intestinal system, Blood)

Specific target organ toxicity - repeated exposure : Category 1 (Nervous system, Bladder, Gastro-intestinal system, Blood)

SAFETY DATA SHEET

MARSHAL 200SC



Version 1.0	Revision Date: 2025/11/21	SDS Number: 50001575	Date of last issue: - Date of first issue: 2025/11/21
----------------	------------------------------	-------------------------	--

Specific target organ toxicity - repeated exposure (Oral) : Category 2 (Kidney)

Short-term (acute) aquatic hazard : Category 1

Long-term (chronic) aquatic hazard : Category 1

GHS label elements

Hazard pictograms : 

Signal Word : DANGER

Hazard Statements : H302 + H332 Harmful if swallowed or if inhaled.
H370 Causes damage to organs (Nervous system, Bladder, Gastro-intestinal system, Blood).
H372 Causes damage to organs (Nervous system, Bladder, Gastro-intestinal system, Blood) through prolonged or repeated exposure.
H373 May cause damage to organs (Kidney) through prolonged or repeated exposure if swallowed.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements : **Prevention:**
P260 Do not breathe mist or vapors.
P264 Wash skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P273 Avoid release to the environment.
Response:
P301 + P312 + P330 IF SWALLOWED: Call a POISON CENTER/ doctor if you feel unwell. Rinse mouth.
P304 + P340 + P312 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER/ doctor if you feel unwell.
P308 + P311 IF exposed or concerned: Call a POISON CENTER/ doctor.
P391 Collect spillage.
Storage:
P405 Store locked up.
Disposal:
P501 Dispose of contents/ container to an approved waste disposal plant.

SAFETY DATA SHEET



MARSHAL 200SC

Version 1.0 Revision Date: 2025/11/21 SDS Number: 50001575 Date of last issue: -
Date of first issue: 2025/11/21

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
carbosulfan (ISO)	55285-14-8	≥ 15 -< 25
Silicic acid, aluminum sodium salt	1344-00-9	≥ 5 -< 7
ethanediol	107-21-1	≥ 3 -< 5
Ethoxylated propoxylated alcohols(C=12-14)	68439-51-0	$\geq 1,5$ -< 2,5
1,2-benzisothiazol-3(2H)-one	2634-33-5	$\geq 0,025$ -< 0,1

4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Show this material safety data sheet to the doctor in attendance.
Do not leave the victim unattended.
- If inhaled : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.
- In case of skin contact : Get medical attention if irritation develops and persists.
Wash off with soap and water.
- In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
Take victim immediately to hospital.
- Most important symptoms and effects, both acute and delayed : Harmful if swallowed or if inhaled.
Causes damage to organs.
Causes damage to organs through prolonged or repeated exposure.
- Notes to physician : It may be helpful to show this safety data sheet to physician.
Treat symptomatically.

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.

SAFETY DATA SHEET



MARSHAL 200SC

Version 1.0	Revision Date: 2025/11/21	SDS Number: 50001575	Date of last issue: - Date of first issue: 2025/11/21
----------------	------------------------------	-------------------------	--

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|--|---|
| Unsuitable extinguishing media | : High volume water jet
Do not spread spilled material with high-pressure water streams. |
| Specific hazards during fire fighting | : Do not allow run-off from fire fighting to enter drains or water courses. |
| Hazardous combustion products | : Carbon oxides
Sulfur oxides
Nitrogen oxides (NOx) |
| Specific extinguishing methods | : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. |
| Special protective equipment for fire-fighters | : Wear self-contained breathing apparatus for firefighting if necessary. |

6. ACCIDENTAL RELEASE MEASURES

- | | |
|---|---|
| Personal precautions, protective equipment and emergency procedures | : Use personal protective equipment.
Ensure adequate ventilation.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material. |
| Environmental precautions | : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities. |
| Methods and materials for containment and cleaning up | : Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).
Keep in suitable, closed containers for disposal. |

7. HANDLING AND STORAGE

- | | |
|---|--|
| Advice on protection against fire and explosion | : Normal measures for preventive fire protection. |
| Advice on safe handling | : Do not breathe vapors/dust.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Dispose of rinse water in accordance with local and national regulations. |
| Conditions for safe storage | : Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage. |

SAFETY DATA SHEET



MARSHAL 200SC

Version 1.0 Revision Date: 2025/11/21 SDS Number: 50001575 Date of last issue: -
Date of first issue: 2025/11/21

Electrical installations / working materials must comply with the technological safety standards.

Further information on storage stability : No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
Silicic acid, aluminum sodium salt	1344-00-9	TWA	2 mg/m3 (Aluminum)	VN OEL
		TWA (Respirable particulate matter)	1 mg/m3 (Aluminum)	ACGIH
ethanediol	107-21-1	TWA (Vapor)	60 mg/m3	VN OEL
		TWA (aerosols, mist)	10 mg/m3	VN OEL
		STEL (Vapor)	125 mg/m3	VN OEL
		STEL (aerosols, mist)	20 mg/m3	VN OEL
		TWA (Vapor) STEL (Vapor) STEL (Inhalable fraction, Aerosol only)	25 ppm 50 ppm 10 mg/m3	ACGIH ACGIH ACGIH

Personal protective equipment

Respiratory protection : In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.

Hand protection

Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Hygiene measures : Wash hands before breaks and at the end of workday.

SAFETY DATA SHEET

MARSHAL 200SC



Version 1.0	Revision Date: 2025/11/21	SDS Number: 50001575	Date of last issue: - Date of first issue: 2025/11/21
----------------	------------------------------	-------------------------	--

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	: liquid
Form	: aqueous suspension concentrate
Color	: red brown
Odor	: mild, phenol-like
pH	: 8 - 9
Melting point	: Not applicable
Boiling point	: No data available
Relative density	: 1,050
Explosive properties	: Not explosive
Oxidizing properties	: Non-oxidizing

10. STABILITY AND REACTIVITY

Reactivity	: No decomposition if stored and applied as directed.
Chemical stability	: No decomposition if stored and applied as directed.
Possibility of hazardous reactions	: No decomposition if stored and applied as directed.
Conditions to avoid	: Protect from frost, heat and sunlight.
Incompatible materials	: Strong bases Strong oxidizing agents Strong acids
Hazardous decomposition products	: Stable under recommended storage conditions.

MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

11. TOXICOLOGICAL INFORMATION**Acute toxicity**

Harmful if swallowed or if inhaled.

Product:

Acute oral toxicity	:	LD50 (Rat): 500 mg/kg
Acute inhalation toxicity	:	LC50 (Rat): 2,27 - 3,28 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403
Acute dermal toxicity	:	LD50 (Rat): > 2.000 mg/kg Method: OECD Test Guideline 402 Assessment: The substance or mixture has no acute dermal toxicity

Components:**carbosulfan (ISO):**

Acute oral toxicity	:	LD50 (Rat, female): 185 mg/kg
Acute inhalation toxicity	:	LC50 (Rat, female): 0,15 mg/l Exposure time: 4 h Test atmosphere: dust/mist
Acute dermal toxicity	:	LD50 (Rat): > 2.000 mg/kg

Silicic acid, aluminum sodium salt:

Acute oral toxicity	:	LD50 (Rat, male and female): 10.000 mg/kg Method: OECD Test Guideline 401 Remarks: Based on data from similar materials
Acute inhalation toxicity	:	LC0 (Rat, male and female): > 2,08 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Remarks: Based on data from similar materials no mortality
Acute dermal toxicity	:	LD50 (Rabbit): > 5.000 mg/kg Method: OECD Test Guideline 402

ethanediol:

Acute inhalation toxicity	:	LC50 (Rat, male and female): > 2,5 mg/l Exposure time: 6 h Test atmosphere: dust/mist Remarks: no mortality
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SAFETY DATA SHEET



MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Acute dermal toxicity : LD50 (Mouse, male and female): > 3.500 mg/kg

Ethoxylated propoxylated alcohols(C=12-14):

Acute oral toxicity : LD50 (Rat): > 2.000 - 5.000 mg/kg
Method: OECD Test Guideline 401

Acute dermal toxicity : LD50 (Rat): > 5.000 mg/kg

1,2-benzisothiazol-3(2H)-one:

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species : Rabbit
Method : OECD Test Guideline 404
Result : No skin irritation

Components:

carbosulfan (ISO):

Species : Rabbit
Result : slight irritation

Silicic acid, aluminum sodium salt:

Species : Rabbit
Result : No skin irritation

ethanediol:

Species : Rabbit
Result : No skin irritation

Ethoxylated propoxylated alcohols(C=12-14):

Species : Rabbit
Method : OECD Test Guideline 404
Result : slight irritation

1,2-benzisothiazol-3(2H)-one:

Species : Rabbit
Exposure time : 72 h
Method : OECD Test Guideline 404
Result : No skin irritation

SAFETY DATA SHEET

MARSHAL 200SC



Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species	:	Rabbit
Result	:	No eye irritation
Method	:	OECD Test Guideline 405

Components:

carbosulfan (ISO):

Species	:	Rabbit
Result	:	slight irritation

Silicic acid, aluminum sodium salt:

Species	:	Rabbit
Result	:	No eye irritation

ethanediol:

Species	:	Rabbit
Result	:	No eye irritation

Ethoxylated propoxylated alcohols(C=12-14):

Species	:	Rabbit
Result	:	No eye irritation
Method	:	Draize Test

1,2-benzisothiazol-3(2H)-one:

Species	:	Bovine cornea
Result	:	No eye irritation
Method	:	OECD Test Guideline 437

Species	:	Rabbit
Result	:	Irreversible effects on the eye
Method	:	EPA OPP 81-4

Respiratory or skin sensitization

Skin sensitization

Based on available data, the classification criteria are not met.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Test Type	:	Buehler Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Not a skin sensitizer.

MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Components:**carbosulfan (ISO):**

Test Type	: Buehler Test
Species	: Guinea pig
Method	: OECD Test Guideline 406
Result	: Not a skin sensitizer.

ethanediol:

Test Type	: Maximization Test
Species	: Guinea pig
Result	: Does not cause skin sensitization.

Ethoxylated propoxylated alcohols(C=12-14):

Test Type	: Maximization Test
Routes of exposure	: Skin contact
Species	: Guinea pig
Method	: OECD Test Guideline 406
Result	: Does not cause skin sensitization.

1,2-benzisothiazol-3(2H)-one:

Test Type	: Maximization Test
Species	: Guinea pig
Method	: OECD Test Guideline 406
Result	: May cause sensitization by skin contact.

Species	: Guinea pig
Method	: FIFRA 81.06
Result	: May cause sensitization by skin contact.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:**carbosulfan (ISO):**

Genotoxicity in vitro	: Test Type: reverse mutation assay Test system: Salmonella typhimurium Result: negative
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Test Type: reverse mutation assay Test system: Escherichia coli Result: negative
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Test Type: gene mutation test Test system: Chinese hamster cells Result: negative

Test Type: Chromosome aberration test in vitro Test system: Chinese hamster cells Result: negative
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MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Genotoxicity in vivo : Test Type: chromosome aberration assay
Species: mice
Result: negative

Silicic acid, aluminum sodium salt:

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: chromosome aberration assay
Species: Rat (male)
Application Route: Oral
Result: negative
Remarks: Based on data from similar materials

ethanediol:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OPPTS 870.5100
Result: negative

Genotoxicity in vivo : Test Type: dominant lethal test
Species: Rat
Application Route: Oral
Result: negative

Ethoxylated propoxylated alcohols(C=12-14):

Genotoxicity in vitro : Test Type: Ames test
Test system: Bacillus subtilis
Method: OECD Test Guideline 471
Result: negative

1,2-benzisothiazol-3(2H)-one:

Genotoxicity in vitro : Test Type: gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative

Test Type: Ames test
Method: OECD Test Guideline 471
Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: positive

Genotoxicity in vivo : Test Type: unscheduled DNA synthesis assay
Species: Rat (male)
Cell type: Liver cells

SAFETY DATA SHEET

MARSHAL 200SC



Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Application Route: Ingestion
Exposure time: 4 h
Method: OECD Test Guideline 486
Result: negative

Test Type: Micronucleus test
Species: Mouse
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

carbosulfan (ISO):

Species : Mouse
Exposure time : 2 Years
NOAEL : 2,5 mg/kg bw/day
Result : negative

Species : Rat
Exposure time : 2 Years
NOAEL : 1 mg/kg bw/day
Result : negative

Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

Silicic acid, aluminum sodium salt:

Species : Rat, male and female
Application Route : Oral
Exposure time : 103 weeks
Result : negative
Remarks : Based on data from similar materials

ethanediol:

Species : Mouse
Application Route : Oral
Exposure time : 24 month(s)
Result : negative

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

carbosulfan (ISO):

SAFETY DATA SHEET



MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Effects on fertility : Test Type: Three-generation study
Species: Rat
Application Route: Oral
General Toxicity Parent: NOAEL: 1,2 mg/kg bw/day
Fertility: NOAEL: 1,2 mg/kg bw/day
Result: negative

Effects on fetal development : Test Type: Embryo-fetal development
Species: Rat
Application Route: Oral
General Toxicity Maternal: NOAEL: 2 mg/kg bw/day
Developmental Toxicity: NOAEL: 2
Result: negative

Test Type: Embryo-fetal development
Species: Rabbit
Application Route: Oral
General Toxicity Maternal: NOAEL: 5 mg/kg bw/day
Developmental Toxicity: NOAEL: 10
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

1,2-benzisothiazol-3(2H)-one:

Effects on fertility : Species: Rat, male
Application Route: Ingestion
General Toxicity Parent: NOAEL: 18,5 mg/kg body weight
General Toxicity F1: NOAEL: 48 mg/kg body weight
Fertility: NOAEL: 112 mg/kg bw/day
Symptoms: No effects on reproduction parameters.
Method: OPPTS 870.3800
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

STOT-single exposure

Causes damage to organs (Nervous system, Bladder, Gastro-intestinal system, Blood).

Components:

carbosulfan (ISO):

Target Organs : Nervous system, Bladder, Gastro-intestinal system, Blood
Assessment : The substance or mixture is classified as specific target organ toxicant, single exposure, category 1.

STOT-repeated exposure

Causes damage to organs (Nervous system, Bladder, Gastro-intestinal system, Blood) through prolonged or repeated exposure.

May cause damage to organs (Kidney) through prolonged or repeated exposure if swallowed.

SAFETY DATA SHEET



MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Components:

carbosulfan (ISO):

Target Organs	:	Nervous system, Bladder, Gastro-intestinal system, Blood
Assessment	:	The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

ethanediol:

Routes of exposure	:	Oral
Target Organs	:	Kidney
Assessment	:	The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.

1,2-benzisothiazol-3(2H)-one:

Assessment	:	The substance or mixture is not classified as specific target organ toxicant, repeated exposure.
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Repeated dose toxicity

Components:

carbosulfan (ISO):

Species	:	Rat
NOAEL	:	2 mg/kg bw/day
Application Route	:	Oral
Exposure time	:	90 days

Species	:	Dog
NOAEL	:	1.6 mg/kg bw/day
Application Route	:	Oral
Exposure time	:	6 months

Silicic acid, aluminum sodium salt:

Species	:	Rat, male and female
NOAEL	:	2.500 - 3.200 mg/kg
Application Route	:	Oral
Exposure time	:	2 years
Remarks	:	Based on data from similar materials

Species	:	Rat, male and female
NOAEL	:	0,0013 mg/l
Application Route	:	Inhalation
Exposure time	:	13 weeks
Remarks	:	Based on data from similar materials

ethanediol:

Species	:	Rat
NOAEL	:	150 mg/kg
Application Route	:	Oral
Exposure time	:	12 Months

SAFETY DATA SHEET



MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Species	: Dog
NOAEL	: > 2.200 - < 4.400 mg/kg
Application Route	: Dermal
Exposure time	: 4 Weeks
Method	: OECD Test Guideline 410

1,2-benzisothiazol-3(2H)-one:

Species	: Rat, male and female
NOAEL	: 15 mg/kg
Application Route	: Ingestion
Exposure time	: 28 d
Method	: OECD Test Guideline 407
Symptoms	: Irritation

Species	: Rat, male and female
NOAEL	: 69 mg/kg
Application Route	: Ingestion
Exposure time	: 90 d
Symptoms	: Irritation, Reduced body weight

Aspiration toxicity

Based on available data, the classification criteria are not met.

Components:

carbosulfan (ISO):

The substance does not have properties associated with aspiration hazard potential.

Further information

Product:

Remarks	: No data available
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12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

carbosulfan (ISO):

Toxicity to fish	: LC50 (Lepomis macrochirus (Bluegill sunfish)): 0,015 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	: EC50 (Daphnia magna (Water flea)): 0,0015 mg/l Exposure time: 48 h
Toxicity to algae/aquatic plants	: EC50 (Pseudokirchneriella subcapitata (microalgae)): > 20 mg/l Exposure time: 96 h
M-Factor (Acute aquatic tox-	: 100

SAFETY DATA SHEET



MARSHAL 200SC

Version 1.0	Revision Date: 2025/11/21	SDS Number: 50001575	Date of last issue: - Date of first issue: 2025/11/21
----------------	------------------------------	-------------------------	--

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Toxicity to fish (Chronic toxicity) : NOEC (Pimephales promelas (fathead minnow)): 0,00828 mg/l
Exposure time: 21 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 0,0032 mg/l
Exposure time: 21 d

M-Factor (Chronic aquatic toxicity) : 10

Toxicity to terrestrial organisms : (Apis mellifera (bees)): 1,035 µg/bee
Remarks: Oral

(Apis mellifera (bees)): 0,18 µg/bee
Remarks: Contact

LD50 (Anas platyrhynchos (Mallard duck)): 10 mg/kg

Silicic acid, aluminum sodium salt:

Toxicity to fish : LL50 (Danio rerio (zebra fish)): 10.000 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): 10.000 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EL50 (Desmodesmus subspicatus (green algae)): 10.000 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

ethanediol:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): > 72.860 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : IC50 (Pseudokirchneriella subcapitata (green algae)): 10.940 mg/l
Exposure time: 96 h

Toxicity to fish (Chronic toxicity) : (Menidia peninsulae (tidewater silverside)): 1.500 mg/l
Exposure time: 28 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : (Daphnia magna (Water flea)): 33.911 mg/l
Exposure time: 21 d

MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Toxicity to microorganisms : (activated sludge): > 1.995 mg/l
Exposure time: 30 min
Method: ISO 8192

Ethoxylated propoxylated alcohols(C=12-14):

Toxicity to fish : LC50 (Leuciscus idus (Golden orfe)): > 1 - 10 mg/l
Exposure time: 48 h
Method: DIN 38412

Toxicity to daphnia and other : EC50 (Daphnia magna (Water flea)): 1 - 10 mg/l
aquatic invertebrates
Exposure time: 24 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic : EC50 (Desmodesmus subspicatus (green algae)): > 1 - 10
plants
mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

EC10 (Desmodesmus subspicatus (green algae)): > 0,1 - 1
mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

1,2-benzisothiazol-3(2H)-one:

Toxicity to fish : LC50 (Cyprinodon variegatus (sheepshead minnow)): 16,7
mg/l
Exposure time: 96 h
Test Type: static test

LC50 (Oncorhynchus mykiss (rainbow trout)): 2,15 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other : EC50 (Daphnia magna (Water flea)): 2,9 mg/l
aquatic invertebrates
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202

Toxicity to algae/aquatic : EC50 (Pseudokirchneriella subcapitata (green algae)): 0,070
plants
mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

NOEC (Pseudokirchneriella subcapitata (green algae)): 0,04
mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

M-Factor (Acute aquatic tox- : 10
icity)

Toxicity to microorganisms : EC50 (activated sludge): 24 mg/l

SAFETY DATA SHEET

MARSHAL 200SC



Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

EC50 (activated sludge): 12,8 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

Persistence and degradability

Components:

carbosulfan (ISO):

Biodegradability : Result: Not readily biodegradable.
Biodegradation: 28 %
Exposure time: 28 d

Stability in water : Remarks: Hydrolyzes readily.

Silicic acid, aluminum sodium salt:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

ethanediol:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 90 - 100 %
Exposure time: 10 d
Method: OECD Test Guideline 301A

Ethoxylated propoxylated alcohols(C=12-14):

Biodegradability : Result: Readily biodegradable.
Biodegradation: > 60 %
Exposure time: 28 d
Method: OECD Test Guideline 301F

1,2-benzisothiazol-3(2H)-one:

Biodegradability : Result: rapidly biodegradable
Method: OECD Test Guideline 301C

Bioaccumulative potential

Components:

carbosulfan (ISO):

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 990
Remarks: Can accumulate in aquatic organisms.

Partition coefficient: n- : log Pow: 5,37

MARSHAL 200SC



octanol/water pH: 8
Method: OECD Test Guideline 107

Partition coefficient: n-octanol/water : Remarks: No data available

Partition coefficient: n-octanol/water : log Pow: -1,36

Bioaccumulation	: Species: <i>Lepomis macrochirus</i> (Bluegill sunfish) Bioconcentration factor (BCF): 6,62 Exposure time: 56 d Method: OECD Test Guideline 305 Remarks: Substance is not persistent, bioaccumulative, and toxic (PBT).
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Partition coefficient: n-octanol/water : log Pow: 0,7 (20 °C)
pH: 7
log Pow: 0,99 (20 °C)
pH: 5

Components:

Distribution among environmental compartments	Remarks: Slightly mobile in soils
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Distribution among environmental compartments	: Koc: 9,33 ml/g, log Koc: 0,97 Method: OECD Test Guideline 121 Remarks: Highly mobile in soils
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Product:

Additional ecological information	: An environmental hazard cannot be excluded in the event of unprofessional handling or disposal. Very toxic to aquatic life with long lasting effects.
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SAFETY DATA SHEET

MARSHAL 200SC



Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.

14. TRANSPORT INFORMATION

International Regulations

UNRTDG

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Carbosulfan)
Class : 9
Packing group : III
Labels : 9
Environmentally hazardous : yes

IATA-DGR

UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s.
(Carbosulfan)
Class : 9
Packing group : III
Labels : Miscellaneous
Packing instruction (cargo aircraft) : 964
Packing instruction (passenger aircraft) : 964
Environmentally hazardous : yes

IMDG-Code

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Carbosulfan)
Class : 9
Packing group : III
Labels : 9
EmS Code : F-A, S-F
Marine pollutant : yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

SAFETY DATA SHEET

MARSHAL 200SC



Version 1.0	Revision Date: 2025/11/21	SDS Number: 50001575	Date of last issue: - Date of first issue: 2025/11/21
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Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Law on Chemicals No. 06/2007/QH12

The ingredients of this product are reported in the following inventories:

TCSI	: Not in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. 2,3-DIHYDRO-2,2-DIMETHYLBENZOFURAN-7-YL (DIBUTYLAMINTHIO)METHYLCARBAMATE PMSIL 30 (PINTU)
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

16. OTHER INFORMATION

Revision Date	: 2025/11/21
Date format	: yyyy/mm/dd

Full text of other abbreviations

ACGIH	: USA. ACGIH Threshold Limit Values (TLV)
VN OEL	: Vietnam. Occupational Exposure Limits

SAFETY DATA SHEET



MARSHAL 200SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/11/21	50001575	Date of first issue: 2025/11/21

ACGIH / TWA	: 8-hour, time-weighted average
ACGIH / STEL	: Short-term exposure limit
VN OEL / TWA	: Time weighted average
VN OEL / STEL	: Short term exposure limit

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECL - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

Disclaimer

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